

Internship report:
Tracking the S-matrix across dimensions, a Regge study

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Abstract

This report outlines the theoretical and computational research conducted during a three-month internship at the Laboratoire d'Annecy de Physique Théorique (LAPTh), under the supervision of Piotr Tourkine. Beginning with a concise introduction to the foundational frameworks of Regge theory, we review the non-perturbative bounds and optimal amplitude spaces established across various spacetime dimensions in [8]. Building directly upon these bootstrap constraints, we present an investigation into the corresponding complex Regge trajectories $J(s)$, tracking their analytic structure via a newly developed numerical continuation pipeline.

Finally, we explore the on-shell residues of the Regge poles within the Veneziano amplitude. This investigation builds upon previous analytical frameworks to track the geometric limit curves along which the corresponding polynomial zeros accumulate. We present the current state of this ongoing work, outlining our partial results within a multi-saddle asymptotic expansion and defining the computational roadmap required to fully resolve these transcendental boundary structures.

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Chapter 1

Introduction

In the 1940s and 1950s, physicists such as Schwinger, Tomonaga, Feynman, and Dyson formulated a successful framework that unified two revolutionary pillars from the turn of the century: special relativity and quantum mechanics. In its standard formulation, the theory starts from a Lagrangian and, rather than yielding physical predictions directly, it follows the multi-step pipeline outlined in Fig. 1.1.

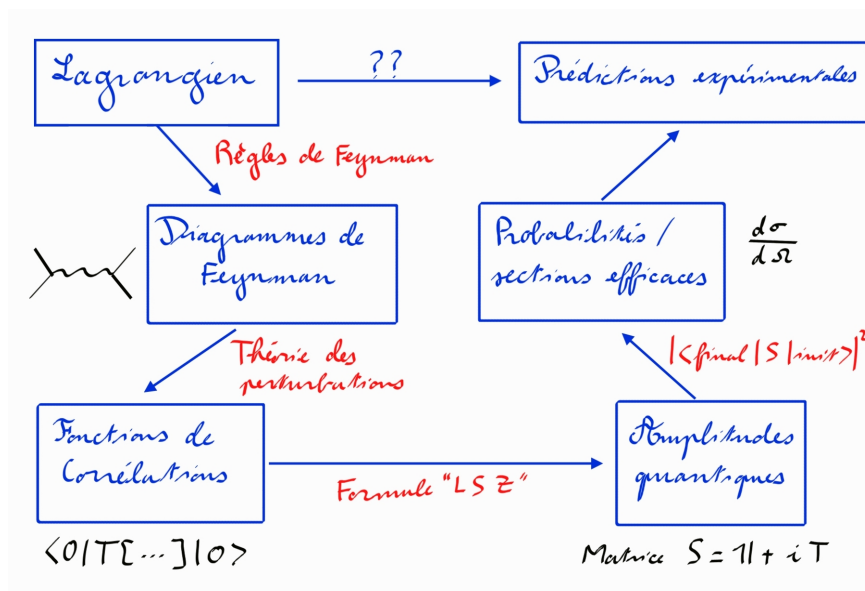


Figure 1.1: Explanation of how perturbative QFT works.

However, in the 1960s, physicists encountered a severe roadblock: the strong interaction possesses a coupling constant that becomes large at hadronic scales, preventing a conventional perturbative treatment. Consequently, the quantum field theoretic framework that had been so successful for quantum electrodynamics could not be readily applied to nuclear forces. Before the advent of quantum chromodynamics (QCD), an alternative path was pursued, inspired by an intuition pioneered by Heisenberg: “In the absence of knowledge of a microscopic theory, let us try to see what constraints the fundamental principles impose. Principles which we know will remain valid even after the discovery of that theory.” This paradigm of constructing and constraining a theory purely through its fundamental physical and mathematical consistency conditions is known as the *bootstrap*.

What general principles guide this bootstrap program? The primary pillar is causality, which is conventionally encoded via the analyticity of the scattering amplitude over specific domains of its complexified kinematic variables. While a comprehensive derivation for the relativistic S -matrix is beyond the scope of this introduction, the underlying mechanism can be illustrated through a simpler, classic example: a linear system governed by a Green's function $G(\omega)$ in frequency space. The temporal response of the system is given by the inverse Fourier transform:

$$G(t) = \int_{-\infty}^{+\infty} \frac{d\omega}{2\pi} G(\omega) e^{-i\omega t}. \quad (1.1)$$

If $G(\omega)$ is analytic in the upper half of the complex ω -plane and satisfies appropriate decay conditions at infinity, the integration path can be closed via a semi-circular contour in the upper half-plane for $t < 0$. By Cauchy's integral theorem, this closed contour integral vanishes identically. Consequently, we obtain $G(t) = 0$ for $t < 0$, which can be compactly written as $G(t) = \Theta(t)G(t)$ (where Θ denotes the Heaviside step function), thereby demonstrating that the system is strictly causal.

A conceptually analogous relation between causality and analyticity applies to the relativistic S -matrix, though it involves significantly greater kinematic complexity; for a comprehensive review, see [14]. In what follows, we adopt the postulate of *maximal analyticity*, assuming that the scattering amplitude is analytic everywhere in the complex domain except for the poles and branch cuts strictly dictated by physical thresholds and unitarity.

A second essential ingredient for the bootstrap is crossing symmetry. To illustrate this, consider a two-to-two ($2 \rightarrow 2$) scattering process. Crossing symmetry states that the amplitude for $12 \rightarrow 34$ is given by the same analytic function describing the crossed-channel processes $1\bar{3} \rightarrow \bar{2}4$ and $1\bar{4} \rightarrow \bar{2}3$.¹ This property holds rigorously in axiomatic QFT, as demonstrated by Bros, Epstein, and Glaser [2].

Adopting the mostly-plus Minkowski metric signature $(-, +, \dots, +)$, we define the standard Mandelstam invariants for $2 \rightarrow 2$ scattering as: $s = -(p_1 + p_2)^2$, $t = -(p_1 - p_3)^2$, and $u = -(p_1 - p_4)^2$, where p_1, p_2 denote the incoming four-momenta and p_3, p_4 represent the outgoing momenta. For identical particles of mass m , these variables satisfy the kinematic identity $s + t + u = 4m^2$. Choosing units where $m = 1$, this relation simplifies to $s + t + u = 4$, effectively eliminating the independent functional reliance on the u -channel. Defining the invariant scattering amplitude $A(s, t)$ via the transition component of the S -matrix (see section 2.2.2), crossing symmetry for identical scalar states reduces to the elegant functional identities:

$$A(s, t) = A(t, s) = A(s, 4 - s - t). \quad (1.2)$$

The final fundamental pillar of the bootstrap program is unitarity, a direct consequence of the probabilistic interpretation of quantum mechanics. To ensure the conservation of total probability under time evolution, the S -matrix—which maps asymptotic *in*-states in the far past to *out*-states in the far future—must be a unitary operator. This physical requirement translates to the operator identity:

$$SS^\dagger = \mathbb{I}. \quad (1.3)$$

Unitarity plays a defining role in constraining the S -matrix bootstrap, a feature we will analyze in detail in Sec. 3.2.

In addition to these principles, several other constraints must be satisfied. First, the S -matrix must respect Poincaré invariance. This space-time symmetry forms the backbone of the rigorous proof for crossing symmetry and will play a pivotal role in our subsequent construction. Furthermore, phenomenological inputs, such as the linear behavior of Regge trajectories, must be properly incorporated.

From these constraints, certain properties of the S -matrix, such as analytical bounds on the amplitude, can be derived. However, the bootstrap conditions are generally underconstrained, preventing a complete

¹ $\bar{i}$ denotes the antiparticle of i .

classification of all physical quantum field theories.² This lack of predictive uniqueness led to the decline of the program in the 1970s.

In recent years, however, the bootstrap has experienced a major revival driven by numerical methods. To illustrate the numerical approach using a toy model, suppose we wish to constrain two physical observables, a and b . The objective is to map the region of the (a, b) parameter space compatible with the fundamental bootstrap constraints. This can be achieved via two complementary strategies. The *primal* bootstrap constructs explicit, valid amplitudes to map the allowed region from the inside out, offering the advantage of yielding a concrete physical amplitude. Conversely, the *dual* bootstrap excludes mathematically incompatible parameter values, bounding the allowed region from the outside in, a bounding technology systematically developed using dual optimization functionals in [10]. This numerical optimization leads to the landscape illustrated in Fig. 1.2, where the domain of physically viable theories is tightly constrained between the primal and dual boundaries.

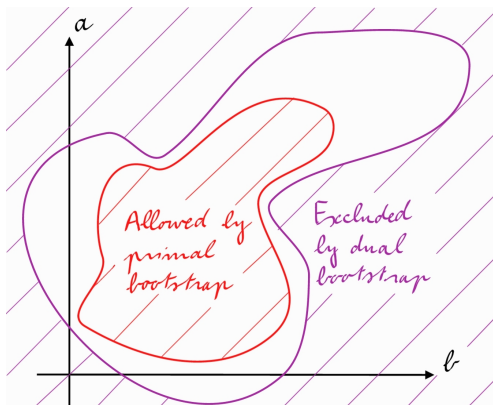


Figure 1.2: Schematic representation of the primal and dual boundaries in the numerical S -matrix bootstrap.

The remainder of this report is structured as follows. We begin with a brief review of the theoretical machinery underlying the S -matrix bootstrap. Our primary analysis builds directly upon the primal numerical framework established by Gumus, Metayer, and Tourkine [8]; their specific ansatz and the relevant physical observables are detailed in Sec. 3.1. After reviewing the landscape of viable scattering amplitudes mapped in [8], we introduce the core computation of this internship: evaluating the partial-wave projection of these numerically optimized amplitudes.

We first demonstrate why a closed-form analytical evaluation of this projection is intractable, thereby motivating the numerical implementation presented in Sec. 4.2. Following this, we outline an efficient method developed to rapidly extract the corresponding Regge trajectories, accompanied by our preliminary numerical results. Finally, Sec. 5.3 is dedicated to a parallel project conducted during this internship: an analytical study of the residues of Regge poles within the classic Veneziano amplitude.

²Such a classification becomes achievable under the severe constraints of a two-dimensional spacetime combined with integrability or conformal symmetry.

Chapter 2

A Minimal Bootstrap Toolkit

2.1 Dispersion Relations

We begin by recalling two classic results from relativistic scattering theory (see, e.g., [3, 21]):

- Across a physical branch cut on the real axis (for a fixed $t_0 \in \mathbb{R}$), real analyticity dictates that the imaginary part of the amplitude equals its discontinuity: $\text{Im}[A(s, t_0)] = \text{Disc}_s[A(s, t_0)]$, where the discontinuity is defined as

$$\text{Disc}_s[A(s, t_0)] = \frac{1}{2i} \lim_{\varepsilon \rightarrow 0^+} [A(s + i\varepsilon, t_0) - A(s - i\varepsilon, t_0)]. \quad (2.1)$$

Under the assumption of maximal analyticity, these kinematic cuts are the only source of discontinuities along the real axis.

- The optical theorem relates the forward scattering amplitude to the total cross-section σ_{tot} via:

$$\sigma_{\text{tot}} = \frac{\text{Im}[A(s, 0)]}{\sqrt{s(s-4)}}. \quad (2.2)$$

Suppose we wish to determine the scattering amplitude at an arbitrary point s in the complex plane for a fixed t without relying on an explicit microscopic theory. We can leverage Cauchy's integral formula from complex analysis:

$$A(s, t) = \frac{1}{2\pi i} \oint_{C_s} \frac{A(s', t)}{s' - s} ds', \quad (2.3)$$

where C_s is a small contour enclosing the point s . Assuming that the amplitude vanishes asymptotically as $|s| \rightarrow \infty$ in any direction,¹ we can deform the integration contour to infinity. In doing so, the boundary at infinity vanishes, and the contour wraps tightly around the non-analytic regions, expressing the amplitude purely in terms of its discontinuities across the physical branch cuts.

By combining unitarity, crossing symmetry, and the contour deformation described above, we can express the scattering amplitude in terms of its imaginary part along the physical thresholds. This yields a dispersion relation of the form:

$$A(s, t) = \frac{1}{\pi} \int_4^\infty ds' \text{Im}[A(s', t)] \left(\frac{1}{s' - s} + \frac{1}{s' - u} \right) \quad (2.4)$$

¹For amplitudes that exhibit polynomial growth at infinity, it is necessary to employ subtracted dispersion relations; see, e.g., [21], Sec. 1.3.2.

This relation forms the basis for constructing self-consistency tests within the bootstrap program. Furthermore, via the optical theorem, experimental total cross-sections can be injected as physical inputs to constrain the low-energy imaginary part of the amplitude, allowing for stable extrapolations.

The primary limitation of this unsubtracted approach arises at high energies, where the integration along the branch cuts can suffer from severe convergence issues. To resolve this behavior, we require an additional framework: Regge theory.

2.2 Regge theory

The core strategy of Regge theory lies in projecting the scattering amplitude into the space of angular momentum. Because the S -matrix is invariant under spatial rotations—which form a subgroup of the Poincaré group—it commutes with the total angular momentum operator and its longitudinal projection: $[S, J^2] = [S, J_z] = 0$. Consequently, the S -matrix can be diagonalized in a simultaneous basis of these conserved generators, giving rise to the standard partial-wave expansion.

The second fundamental insight of the framework is to analytically continue the discrete partial-wave amplitudes $f_J(s)$ to a continuous, complex angular momentum variable J . Although this construction can be generalized to particles carrying arbitrary helicity via the Jacob-Wick formalism, we restrict our exposition to the scattering of spinless (spin-0) particles. The profound physical utility of this complexification will be developed in Sec. 2.2.3.

2.2.1 The Partial-Wave Expansion and Multi-Dimensional Generalization

We consider the $d = 4$ case in this subsection. Let us define the invariant scattering amplitude $A(s, t)$ as the matrix element of the interacting part of the S -matrix, T , between asymptotic two-particle states in the center-of-mass frame: $A(s, t) = \langle p, \theta, \phi | T | p, 0, 0 \rangle$, where the kinematics follow the configuration illustrated in Fig. 2.1.

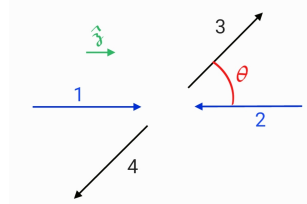


Figure 2.1: Diagram for the 2 to 2 scattering.

To exploit the rotational symmetry of the system, we decompose the continuous momentum states into a basis of definite total angular momentum $|p, J, M\rangle$. The spherical harmonics $Y_{J,M}(\theta, \phi)$ form an orthonormal basis for the Hilbert space $\mathcal{L}^2(S^2)$ over the two-sphere, which realizes the irreducible representations of the rotation group $SO(3)$. By the spectral decomposition of the Laplace-Beltrami operator on S^2 , the transformation between the configuration and angular momentum representations is given exactly by:

$$|p, \theta, \phi\rangle = \sum_{J=0}^{\infty} \sum_{M=-J}^J |p, J, M\rangle Y_{J,M}^*(\theta, \phi). \quad (2.5)$$

By inserting this completeness relation into the matrix element and invoking the rotational invariance of the T -operator ($[T, \mathbf{J}] = 0$), Schur's lemma dictates that the matrix elements are diagonal in J and independent of the magnetic projection quantum number M . We adopt the standard normalization convention:

$$\langle p, J, M | T | p, J', M' \rangle = 64\pi^2 \delta_{J,J'} \delta_{M,M'} f_J(s). \quad (2.6)$$

Furthermore, because the initial state $|p, 0, 0\rangle$ is collinear with the quantization z -axis, it possesses azimuthal symmetry, restricting its projection to $M' = 0$. Evaluating the final-state transformation yields:

$$\begin{aligned} A(s, t) &= \sum_{J, M} \sum_{J'} Y_{J, M}(\theta, \phi) Y_{J', 0}^*(0, 0) \langle p, J, M | T | p, J', 0 \rangle \\ &= 64\pi^2 \sum_{J=0}^{\infty} Y_{J, 0}(\theta, \phi) Y_{J, 0}^*(0, 0) f_J(s). \end{aligned} \quad (2.7)$$

Recalling that $Y_{J, 0}(0, 0) = \sqrt{\frac{2J+1}{4\pi}}$ and $Y_{J, 0}(\theta, \phi) = \sqrt{\frac{2J+1}{4\pi}} P_J(\cos \theta)$, where P_J denotes the standard Legendre polynomial, the expansion collapses to the classic partial-wave series:

$$A(s, t) = 16\pi \sum_{J=0}^{\infty} (2J+1) f_J(s) P_J(z), \quad (2.8)$$

where $z \equiv \cos \theta = 1 + \frac{2t}{s-4}$.²

This framework can be generalized to an arbitrary spacetime dimension $d \geq 4$ by replacing the $SO(3)$ spherical harmonics with the irreducible representations of the little group $SO(d-1)$ governing the center-of-mass angular configurations [5]. The corresponding generalized Legendre polynomials (or ultraspherical/Gegenbauer polynomials normalized to unity at $z = 1$) are given by the hypergeometric profile:

$$P_J^{(d)}(z) = {}_2F_1 \left(-J, J+d-3; \frac{d-2}{2}; \frac{1-z}{2} \right). \quad (2.9)$$

Utilizing the orthogonality relation on the interval $z \in [-1, 1]$,

$$\frac{1}{2} \int_{-1}^1 dz (1-z^2)^{\frac{d-4}{2}} P_J^{(d)}(z) P_{J'}^{(d)}(z) = \frac{\delta_{J, J'}}{\mathcal{N}_d n_J^{(d)}}, \quad (2.10)$$

where the kinematic normalization factor is defined as

$$n_J^{(d)} = \frac{(4\pi)^{\frac{d}{2}} (d+2J-3) \Gamma(d+J-3)}{\pi \Gamma\left(\frac{d-2}{2}\right) \Gamma(J+1)}, \quad (2.11)$$

we can project out individual partial waves from the full amplitude:

$$f_J(s) = \frac{\mathcal{N}_d}{2} \int_{-1}^1 dz (1-z^2)^{\frac{d-4}{2}} P_J^{(d)}(z) A(s, t(s, z)), \quad \text{with } \mathcal{N}_d = \frac{(16\pi)^{\frac{2-d}{2}}}{\Gamma\left(\frac{d-2}{2}\right)}. \quad (2.12)$$

While Eq. (2.12) is mathematically complete for discrete values of $J \in \mathbb{N}$, it cannot be consistently extended to the complex J -plane due to exponential divergence issues at infinity. Overcoming this barrier requires the introduction of the Froissart-Gribov projection.

2.2.2 Froissart-Gribov projection

Prior to deriving the full Froissart-Gribov projection, we must introduce the generalized Legendre function of the second kind, $Q_J^{(d)}(z)$. For an arbitrary spacetime dimension $d \geq 4$, it is defined outside the physical region via the hypergeometric series:

$$Q_J^{(d)}(z) = \frac{c_J^{(d)}}{z^{J+d-3}} {}_2F_1 \left(\frac{J+d-3}{2}, \frac{J+d-2}{2}; J + \frac{d-1}{2}; \frac{1}{z^2} \right), \quad (2.13)$$

²For a comprehensive treatment of the underlying Lie group representations, see [20], Chapter 7.

where the normalization constant $c_J^{(d)}$ reads

$$c_J^{(d)} = \frac{\sqrt{\pi} \Gamma(J+1) \Gamma\left(\frac{d-2}{2}\right)}{2^{J+1} \Gamma\left(J + \frac{d-1}{2}\right)}. \quad (2.14)$$

The function $Q_J^{(d)}(z)$ possesses a branch cut along the real interval $z \in [-1, 1]$. Its discontinuity across this segment maps directly onto the generalized Legendre polynomial $P_J^{(d)}(z)$ through the identity:

$$\text{Disc}_z \left[(z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) \right] = -\frac{\pi}{2} (1 - z^2)^{\frac{d-4}{2}} P_J^{(d)}(z), \quad z \in [-1, 1]. \quad (2.15)$$

This relation can be made transparent by expressing $(z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z)$ as a Cauchy-type integral over the physical discontinuity:³

$$(z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) = \frac{1}{2} \int_{-1}^1 dz' (1 - z'^2)^{\frac{d-4}{2}} \frac{P_J^{(d)}(z')}{z - z'}. \quad (2.16)$$

By utilizing Eq. (2.15), the real boundary integral over the physical region $z \in [-1, 1]$ in Eq. (2.12) can be recast as a closed contour integral. Deforming the path to tightly enclose the segment $[-1, 1]$ in a counter-clockwise direction yields:

$$f_J(s) = -\frac{\mathcal{N}_d}{2\pi i} \oint_{[-1,1]} dz (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) A(s, t(s, z)). \quad (2.17)$$

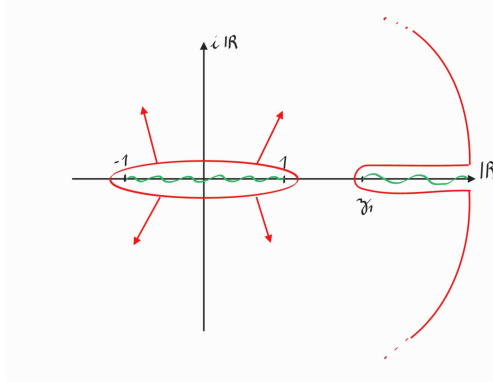


Figure 2.2: Contour for the Froissart-Gribov projection.

We can now deform the integration contour to infinity (see Fig. 2.2), a maneuver justified by the known asymptotic behavior of the generalized Legendre function, $Q_J^{(d)}(z) \sim z^{-J-d+3}$ as $z \rightarrow \infty$, in conjunction with the power-counting constraints imposed by the Froissart bound [7]. In doing so, the boundary term at infinity vanishes identically for sufficiently large $\text{Re}(J)$, and the contour traps the discontinuities across the crossed-channel branch cuts. Exploiting crossing symmetry to map the left- and right-hand cuts into a single localized domain, we arrive at the Froissart-Gribov representation of the partial-wave amplitude:

$$f_J(s) = \frac{2\mathcal{N}_d}{\pi} \int_{z_1}^{+\infty} dz (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) \text{Disc}_t [A(s, t(s, z))], \quad (2.18)$$

³Equivalently, applying the standard distributional relation $\text{Disc}_z [(z' - z)^{-1}] = -\pi \delta(z' - z)$ to Eq. (2.16) recovers the local discontinuity given in Eq. (2.15).

where $z_1 = \frac{s+4}{s-4}$ denotes the position of the physical threshold in the crossed channel.

While this substitution may initially appear to merely exchange one real boundary integral for another without immediate simplification for physical integers $J \in \mathbb{N}$, its profound utility lies in the arena of complexified angular momentum. Unlike the original projection formula, Eq. (2.18) possesses a well-defined, unique analytic continuation to the complex J -plane. Although infinite continuous functions can interpolatively match a discrete set of points, Carlson's theorem [9] guarantees that Eq. (2.18) defines the unique analytic continuation of the partial waves, provided the function satisfies the requisite exponential bounds at infinity in the right half-plane.

2.2.3 And back to momentum space

To extract high-energy physical approximations, we must project our representation back into momentum space. Rather than evaluating the discrete summation in Eq. (2.8) directly—which converges poorly at high energies and obscures the underlying asymptotic dynamics—we employ the Sommerfeld-Watson transform. The first step consists of recasting the discrete partial-wave series as a contour integral in the complex angular momentum plane. By exploiting the fact that the function $1/\sin(\pi J)$ features simple poles at all integers $J \in \mathbb{Z}$ with residues $(-1)^J/\pi$, and noting that $P_J(-z) = (-1)^J P_J(z)$ for $J \in \mathbb{N}_0$, we can express the scattering amplitude as:

$$A(s, t) = -\frac{16\pi}{2i} \int_{C_0} dJ (2J+1) f_J(s) \frac{P_J(-z)}{\sin(\pi J)}, \quad (2.19)$$

where C_0 denotes the contour wrapping tightly around the positive real axis in a clockwise direction, thereby enclosing only the poles located at non-negative integers $J = 0, 1, 2, \dots$ (indicated as the green contour in Fig. 2.3).

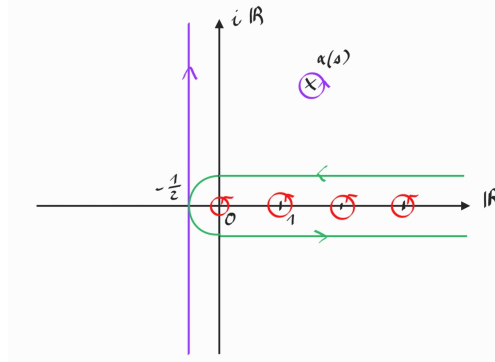


Figure 2.3: Consecutive contours during the derivation of the Sommerfeld-Watson transformation.

We can now "open" the integration contour by deforming C_0 into the left half-plane until it rests along the vertical axis. In doing so, the contour captures the singularities of the partial-wave amplitude $f_J(s)$ located in the right half-plane. Assuming $f_J(s)$ is a meromorphic function of the complex angular momentum J , these singularities manifest as a discrete set of moving poles—known as Regge poles—parameterized near the pole position $\alpha_i(s)$ by:

$$f_J(s) \approx \frac{\beta_i(s)}{J - \alpha_i(s)}, \quad (2.20)$$

where $\beta_i(s)$ represents the residue function.

This deformation leaves a background integral along the line $\text{Re}(J) = -\frac{1}{2}$, a position chosen because it minimizes the asymptotic high-energy contribution of the background background term [3]. Collecting the

residue contributions via Cauchy's theorem yields the full amplitude:

$$A(s, t) = -16\pi^2 \sum_i (2\alpha_i(s) + 1) \beta_i(s) \frac{P_{\alpha_i(s)}(-z)}{\sin(\pi\alpha_i(s))} - \frac{16\pi}{2i} \int_{-\frac{1}{2}-i\infty}^{-\frac{1}{2}+i\infty} dJ (2J+1) f_J(s) \frac{P_J(-z)}{\sin(\pi J)}. \quad (2.21)$$

In the high-energy asymptotic limit, the background integral becomes negligible, leaving the amplitude dominated by the leading Regge poles. This representation yields a clear physical picture when a Regge trajectory passes near a non-negative integer, $\alpha_i(s = m_n^2) = n \in \mathbb{N}_0$. Expanding the sine function and the Legendre function near this threshold isolates a single resonant contribution:

$$A(s, t) \approx 16\pi(2n+1) \beta_i(m_n^2) \frac{P_n(z)}{n - \alpha_i(s)}. \quad (2.22)$$

This expression is functionally identical to the standard Breit-Wigner formula for the exchange of a physical particle of mass m_n and stable spin n . The Sommerfeld-Watson transform thus establishes a direct bridge between poles in the complex angular momentum plane and the particle spectrum of a quantum field theory.

This continuous travel back and forth between momentum space and partial waves allows us to extract clear physical information about asymptotic scattering, providing an elegant basis for implementing a consistent bootstrap setup [21].

Chapter 3

Tracking the S-matrix across dimensions

In this chapter, we review the core structural aspects of the framework introduced in [8], which serves as the foundation for the work conducted during this internship. To maintain a concise exposition, we focus primarily on the technical developments essential to our results; an exhaustive treatment and comprehensive references can be found in the original literature.

3.1 The Ansatz

To systematically explore the space of scattering amplitudes, one must construct a functional ansatz that explicitly incorporates the fundamental principles of the S -matrix bootstrap. A primary constraint is macro-causality, which manifests as analyticity in the complex Mandelstam planes. Unlike a simple polynomial expansion which yields entire functions, physical unitarity in a massive theory dictates that the amplitude possesses distinct non-analytic structures. Above the two-particle threshold $s = 4$, production mechanisms induce branch cuts running from 4 to infinity in the respective Mandelstam channels.

A powerful method to incorporate these branch cuts while preserving a controlled, convergent series expansion is to perform a conformal mapping of the cut complex plane onto the unit disk. This is achieved via the ρ -variable transformation:

$$\rho(s, \sigma) = \frac{\sqrt{\sigma - 4} - \sqrt{4 - s}}{\sqrt{\sigma - 4} + \sqrt{4 - s}}, \quad (3.1)$$

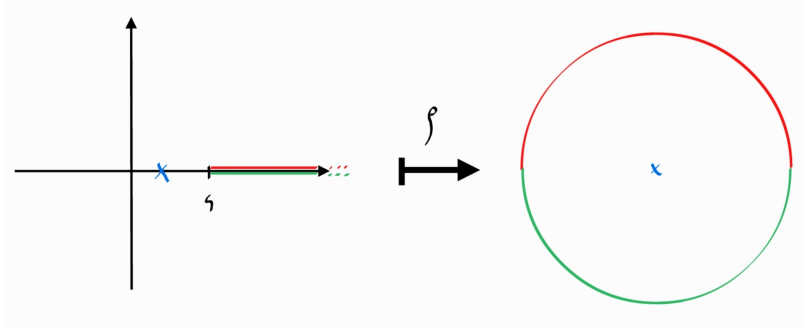


Figure 3.1: Sketch of the conformal map $\rho(s, \sigma)$ for an arbitrary real parameter $\sigma > 4$.

Here, $s \in \mathbb{C}$ and the parameter is chosen within the range $\sigma \in (4, +\infty)$. A schematic behavior of this mapping is illustrated in Fig. 3.1. At this stage, two distinct structural strategies can be pursued to build the full amplitude basis.

The first approach relies on a single fixed value of σ to construct uniform crossing-symmetric polynomial terms, such as $\rho(s, \sigma)^n \rho(t, \sigma)^k \rho(u, \sigma)^l$ for $n, k, l \in \mathbb{N}_0$. For instance, choosing $\sigma = \frac{20}{3}$ maps the fully crossing-symmetric point $s = t = u = \frac{4}{3}$ directly to the origin ($\rho = 0$) of the unit disk. This framework is highly efficient for probing physical scattering regimes well away from the multi-particle thresholds.

Alternatively, one can introduce a discrete set of varying parameters σ_i . This choice effectively generates localized "wavelets" at distinct energy scales, accelerating series convergence when the algorithm scans over a vast energy range. Furthermore, this multi-scale parameterization enables dense numerical meshing and finer resolution near critical kinematic boundaries, such as production thresholds. This latter multi-pole wavelet architecture is the methodology adopted in [8].

To ensure manifest crossing symmetry, the ansatz explicitly sums over all permutations of the Mandelstam variables for each basis term. Furthermore, as shown in [8], elastic unitarity constraints dictate the maximal singular behavior allowed at the two-particle threshold. In $d = 4$, the amplitude can exhibit a threshold divergence scaling as $(4 - s)^{-1/2}$. For higher even spacetime dimensions d , this singular behavior generalizes to:

$$A(s, t) \sim \frac{1}{(4 - s)^{\frac{d-3}{2}}}. \quad (3.2)$$

For odd dimensions, this power-law behavior is accompanied by a mandatory logarithmic correction (the explicit construction of the corresponding numerators and dimensional continuations are detailed in Appendix B of the reference).

To absorb these non-analyticities directly and maximize numerical efficiency, a dedicated threshold term $A_{\text{thresh}}(s, t)$ is factored into the ansatz. This subtraction prevents the optimization algorithm from spending numerical resources trying to reproduce singular local behavior using a smooth series expansion. Combining these components, the full ansatz is structured as:

$$A(s, t) = A_{\text{thresh}}(s, t) + \frac{n_d}{6} \sum_{i,j=1}^{N_{\text{max}}} \sum_{k,l \in \{0,1\}} \alpha_{ijkl} (\rho(s, \sigma_i)^k \rho(t, \sigma_j)^l + \text{perms}(s, t, u)) \delta_{ijkl}, \quad (3.3)$$

where α_{ijkl} represent the free parameters tuned during numerical optimization, n_d is a dimension-dependent normalization constant, and δ_{ijkl} is a structural selection rule governing the allowed index combinations (see Appendix C of the article).

3.2 Physical Quantities and Unitarity Bounds

The low-energy observables of interest in this framework are the low-energy Wilson coefficients parameterizing the scattering amplitude. We focus on the values of the amplitude and its derivatives evaluated at the fully crossing-symmetric subtraction point $s = t = u = \frac{4}{3}$:

$$c_0 = A\left(\frac{4}{3}, \frac{4}{3}\right), \quad (3.4)$$

$$c_2 = \frac{1}{4} \frac{\partial^2}{\partial s^2} A\left(s, \frac{4}{3}\right) \Big|_{s=\frac{4}{3}}. \quad (3.5)$$

These parameters appear directly in the low-energy effective field theory (EFT) expansion around the symmetric point, defined via $A_{\text{eff}}(s, t) = c_0 + c_2(\bar{s}^2 + \bar{t}^2 + \bar{u}^2) + \dots$, where $\bar{x} \equiv x - \frac{4}{3}$. For instance, in a weakly coupled $\lambda\phi^4$ toy model in $d = 4$, c_0 matches the tree-level coupling λ . A rigorous tracking of these coefficients in a more physical setting can be found in [11].

To map the landscape of consistent quantum field theories, we look for the geometrically allowed boundaries of the (c_0, c_2) parameter space. While analyticity and crossing symmetry are structural features built directly into our conformal ansatz, non-perturbative unitarity must be enforced as an external constraint. This condition is implemented in the partial-wave basis, where the S -matrix elements must satisfy:

$$|S_J(s)|^2 \leq 1, \quad \text{with} \quad S_J(s) = 1 + i \frac{(s-4)^{\frac{d-3}{2}}}{\sqrt{s}} f_J(s) \quad (\forall s > 4). \quad (3.6)$$

Furthermore, in the elastic window below the four-particle production threshold ($4 < s < 16$), the constraint sharpens to strict saturation, $|S_J(s)|^2 = 1$.

Expressed as a set of non-linear constraints across continuous energy scales and infinite spins J , this setup constitutes a semi-definite programming (SDP) problem. For a fixed spacetime dimension d , the algorithm delineates a compact region of allowed theories, typically yielding a characteristic "almond-shaped" boundary in the parameter space (see Fig. 3.2).

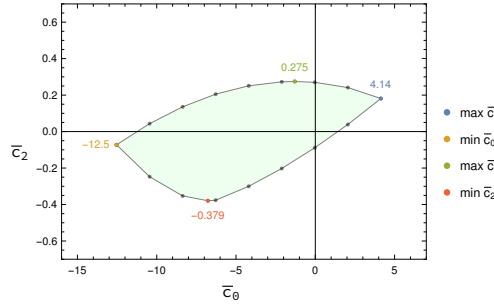


Figure 3.2: Typical bounded region of consistent low-energy EFT parameters obtained via numerical semidefinite programming.

3.3 Results and Open Questions

The primary numerical results established in [8] are summarized by the extremal primal limit points in Fig. 3.3, which trace the boundaries of the allowed low-energy EFT parameter space across discrete sampled values of the spacetime dimension d . Crucially, the geometry of these bounds undergoes sharp qualitative transitions—manifesting as distinctive "kinks"—near $d = 5$ and $d = 7$. While the literature provides preliminary interpretations of these features, the underlying microphysical origin governing these sudden dimensional regime changes remains an open question.

An examination of the complex angular momentum spectrum for the extremal amplitudes maximizing these bounds—accessible via the [SDPBWrapDisplay](#) interactive visualization platform—reveals that the corresponding Regge trajectory structure becomes remarkably intricate in the vicinity of these kinks. Deconstructing this behavior provides a diagnostic tool for identifying the structural features of the underlying UV completion. Specifically, a highly dense rearrangement of Regge poles can hint at the emergence of non-local dynamics or the necessity of incorporating gravitational interactions, which were omitted from this localized flat-space setup. The precise characterization of this complex J -plane topology motivates the core focus of this internship and forms the basis of the results detailed in the subsequent chapters.

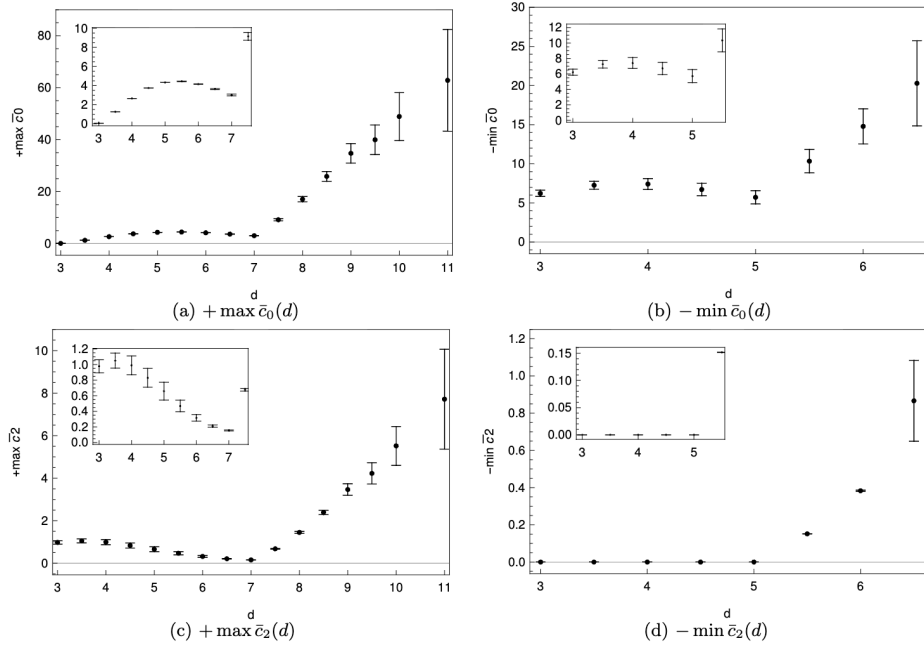


Figure 3.3: Evolution of the low-energy parameter bounds as a function of the spacetime dimension d , showing clear geometric transitions at specific dimensions.

Chapter 4

The Partial-Wave Projection

As highlighted in the previous chapter, the numerical boundaries established in [8] provide rich data while simultaneously raising deep physical questions: What is the exact structural origin of the kinks? What unique dynamics govern the different dimensional regimes?

A powerful approach to decoding these results is to analyze the amplitude's behavior in the complex angular momentum J -plane, tracking the trajectories of the Regge poles. To achieve this, we map our conformal ansatz back into the complex spin plane using the Froissart-Gribov projection introduced in Section 2.2.2. This chapter first details an analytical attempt to evaluate these projection integrals. We then outline the technical limitations that necessitated a transition to a robust numerical framework, describing its implementation and the specialized algorithmic solutions developed during this internship.

4.1 An Attempt at an Analytical Projection

When applying the Froissart-Gribov projection to the conformal ansatz in Eq. (3.3), the problem reduces to a small set of master integrals. While the threshold terms evaluate to closed-form expressions using symbolic computation software such as `Mathematica`, the presence of the conformal ρ -variables introduces two highly non-trivial master integrals:

$$I_t^Q(J, s) = \int_{z_1}^{+\infty} dz (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) \rho_t(t, \sigma), \quad (4.1)$$

$$I_{tu}^Q(J, s) = \int_{z_1}^{+\infty} dz (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) \rho_t(t, \sigma) \rho(u, \tilde{\sigma}), \quad (4.2)$$

The term $\rho_t(t, \sigma)$ represents the directional discontinuity of the conformal wavelets across the physical t -channel cut, defined via:

$$\rho_t(t, \sigma) \equiv \frac{1}{2i} \left(\lim_{\epsilon \rightarrow 0^+} \rho(t + i\epsilon, \sigma) - \lim_{\epsilon \rightarrow 0^-} \rho(t - i\epsilon, \sigma) \right) = \frac{2\sqrt{t-4}\sqrt{\sigma-4}}{t + \sigma - 8}. \quad (4.3)$$

I will focus on the first one which is slightly easier. A good variety of representations of this integral were found during the beginning of my internship. I tried to change variables in order to remove the square root and use methods from [16] but some branch cut polluted the complex plane. I also tried to expand the ρ function and solve order by order but no re-summation was found. The further I got was doing the following:

- We employ the hypergeometric representation of the d -dimensional Legendre function of the second

kind via [DLMF 15.8.12](#) to map the kernel onto a standard Gauss hypergeometric function:

$$(z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) \propto \frac{1}{z^{J+1}} {}_2F_1\left(\frac{J+1}{2}, \frac{J+2}{2}; J + \frac{d+1}{2}; \frac{1}{z^2}\right). \quad (4.4)$$

- We perform a kinematic change of variable from the scattering angle cosine z to a dimensionless parameter w , defined by scaling the t -channel distance above threshold: $t = 4 + (\sigma - 4)w$. Under this transformation, the master integral maps to the semi-infinite domain:

$$I_t^Q(J, s) \propto \int_0^{+\infty} dw \frac{1}{z(w)^{J+1}} {}_2F_1\left(\frac{J+1}{2}, \frac{J+2}{2}; J + \frac{d+1}{2}; \frac{1}{z(w)^2}\right) \frac{\sqrt{w}}{w+1}. \quad (4.5)$$

- Expanding the hypergeometric series term-by-term and interchanging the order of summation and integration allows us to evaluate the remaining sub-integrals via [DLMF 14.6.2](#). This directly yields the closed-form representation up to an overall kinematic prefactor:

$$I_t^Q(J, s) \propto \sum_{n=0}^{\infty} \frac{\Gamma\left(\frac{1}{2}\right) \left(\frac{J+1}{2}\right)_n \left(\frac{J+2}{2}\right)_n \Gamma\left(J + 2n - \frac{1}{2}\right) {}_2F_1\left(J + 2n + 1, \frac{3}{2}; J + 2n + 2; 1 - \frac{2(\sigma-4)}{s+4}\right)}{\Gamma(J + 2n) n! \left(J + \frac{d-1}{2}\right)_n}. \quad (4.6)$$

This infinite series can be formally mapped onto generalized Appell series or Kampé de Fériet hypergeometric functions [\[15\]](#). While the structural and asymptotic properties of these multi-variable hypergeometric functions are thoroughly documented, their real-axis numerical evaluation relies either on slowly convergent series expansions or expensive direct numerical quadratures. Consequently, this analytical expression offers no practical computational advantage for mapping the continuous Regge spectrum, prompting us to pivot to a dedicated, high-precision numerical integration scheme.

4.2 The Numerical Scheme

The numerical methodology implemented in [\[8\]](#) evaluates the partial-wave amplitudes directly from the standard projection. For physical integer spins $J \in \mathbb{N}$ and $\Re(s) > 0$, this representation benefits from a compact integration domain and a completely non-singular integrand. However, this direct projection is ill-defined for complex angular momentum $J \in \mathbb{C}$, see [Section 2.2.2](#). To systematically map the Regge trajectories, we must instead employ the Froissart-Gribov projection.

4.2.1 Constructing the Discontinuity

To evaluate the Froissart-Gribov projection numerically, we must first compute the directional discontinuity of the amplitude across its physical cuts. While the discontinuity of individual wavelet components can be evaluated analytically (as shown in [Section 4.1](#)), extracting the discontinuity of the full optimized ansatz is handled more robustly via a finite- ε numerical prescription. To quantify the systematic error introduced by a finite regulator, we define the regularized discontinuity profile as:

$$\rho_t^\varepsilon(t, \sigma) \equiv \frac{1}{2i} (\rho(t + i\varepsilon, \sigma) - \rho(t - i\varepsilon, \sigma)), \quad (4.7)$$

yielding the regularized projection integral:

$$I_t^\varepsilon(J, s) = \int_{z_1}^{+\infty} dz (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) \rho_t^\varepsilon(t(z), \sigma). \quad (4.8)$$

The regulator $\varepsilon > 0$ must be calibrated to ensure that the truncation error remains negligible when summing the higher-order terms of the full ansatz. We map the parameter space by tracking the relative error metric:

$$\text{Error}_{\text{disc}} = \left| \frac{I_t^\varepsilon(J, s) - I_t(J, s)}{I_t^\varepsilon(J, s) + I_t(J, s)} \right| \quad (4.9)$$

for a wide sampling of kinematic regimes $s \in (4, +\infty)$ and complex spins $J \in \mathbb{C}$, selecting an optimal ε accordingly.

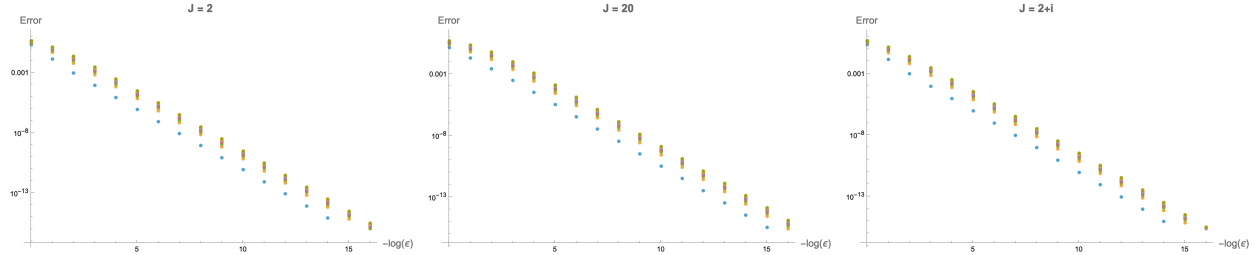


Figure 4.1: The relative error $\text{Error}_{\text{disc}}$ as a function of $-\log_{10} \varepsilon$ for three distinct values of the spin J , evaluated across ten sampled values of the energy scale $s \in [5, 95]$.

As illustrated in Fig. 4.1, accounting for the cumulative error generated by summing several hundred independent basis integrals within the complete ansatz, a strict regulation of the numerical boundary is required. To ensure that the global systematic tolerance remains strictly below 10^{-6} , we select an optimal regulator value of $\varepsilon = 10^{-10}$. This value is maintained uniformly throughout the remainder of the numerical computations.

4.2.2 Asymptotic Tail Subtraction

With the optimal numerical regulator ε established, we can proceed to evaluate the complete projection integral. A direct numerical calculation of the full amplitude ansatz over the semi-infinite domain, however, introduces an immediate computational bottleneck, requiring roughly 30 seconds per evaluation point. This high operational cost stems directly from the non-compact integration domain $[z_1, +\infty)$. Because the integrand possesses a slowly decaying power-law tail, standard adaptive quadrature routines struggle, demanding an excessively dense sampling of function evaluations. Attempting to compactify the domain via coordinate substitutions merely shifts the difficulty, mapping the power-law tail into highly oscillatory phases that yield minimal practical performance gains.

To resolve this issue, we implement an analytical tail subtraction scheme. In the deep asymptotic forward regime ($z \gg 1$), the wavelet discontinuity $\rho_t(t, \sigma)$ scales at worst like $t^{-1/2} \propto z^{-1/2}$. Consequently, the far-forward contribution beyond a sufficiently large cutoff $z_{\text{tail}} \gg 1$ can be mapped onto a known master integral and evaluated analytically:

$$\begin{aligned} \int_{z_{\text{tail}}}^{+\infty} dz (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) z^{-1/2} &= \sqrt{\pi} 2^{-J-2} \Gamma\left(\frac{d}{2} - 1\right) \left(\frac{1}{z_{\text{tail}}^2}\right)^{\frac{J}{2} + \frac{1}{4}} \Gamma\left(\frac{J}{2} + \frac{1}{4}\right) \Gamma(J+1) \\ &\times {}_3\tilde{F}_2\left(\frac{2J+1}{4}, \frac{J+1}{2}, \frac{J+2}{2}; \frac{2J+5}{4}, J + \frac{d-1}{2}; \frac{1}{z_{\text{tail}}^2}\right), \end{aligned} \quad (4.10)$$

where ${}_3\tilde{F}_2$ denotes the regularized generalized hypergeometric function defined in [DLMF 16.2.5](#). Rather than executing an expensive, exact symbolic extraction of the asymptotic coefficient for each of the several hundred individual terms in our ansatz, we perform a localized numerical power-law fit of the total tail profile at $z \approx z_{\text{tail}}$, as illustrated in Fig. 4.2. This hybrid semi-analytical subtraction drastically accelerates the computational routine, isolating the slow numerical quadrature to the compact core domain $[z_1, z_{\text{tail}}]$.

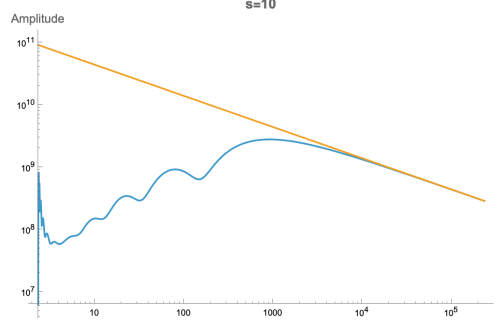


Figure 4.2: Numerical evaluation of the t -channel discontinuity $\text{Disc}_t A(s, z)$ for $d = 6$ at maximum optimization capacity $c_{\max}$ with a truncation cutoff $N_{\max} = 20$ (blue), compared against the asymptotic power-law fit $\alpha_{\text{tail}} z^{-1/2}$ (orange).

As shown in Fig. 4.2, the power-law approximation becomes highly accurate in the far-forward regime, tracking the true discontinuity closely above $z \sim 10^4 z_1$, where $z_1 = 1 + \frac{8}{s-4}$ marks the physical threshold boundary of the Froissart-Gribov integration domain in Eq. (2.18). Denoting the fitted scaling amplitude as α_{tail} , we isolate the analytical tail contribution and evaluate the true numerical residual via:

$$I_{\text{tail}}(z_{\text{tail}}) = \alpha_{\text{tail}} \int_{z_{\text{tail}}}^{+\infty} dz (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) z^{-1/2}, \quad (4.11)$$

$$I_{\text{tail}}^{\text{exact}}(z_{\text{tail}}) = \int_{z_{\text{tail}}}^{+\infty} dz (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) \text{Disc}_t A(s, t(z)). \quad (4.12)$$

To establish a rigorous bound on the systematic error introduced by this asymptotic matching framework, we compute the relative error normalized by the full partial-wave amplitude $f_J(s)$:

$$\text{Error}_{\text{tail}}(z_{\text{tail}}) = \left| \frac{I_{\text{tail}}(z_{\text{tail}}) - I_{\text{tail}}^{\text{exact}}(z_{\text{tail}})}{f_J(s)} \right|. \quad (4.13)$$

Our goal is to identify a sufficiently conservative cutoff scale z_{tail} such that this truncation error remains strictly bounded below 10^{-6} . Scaling the integration domain bounds as illustrated in Fig. 4.3, we observe that setting $z_{\text{tail}} = 50,000 z_1$ satisfies this global precision constraint across all tested kinematic configurations. This threshold is consistently implemented across all subsequent projection calculations.

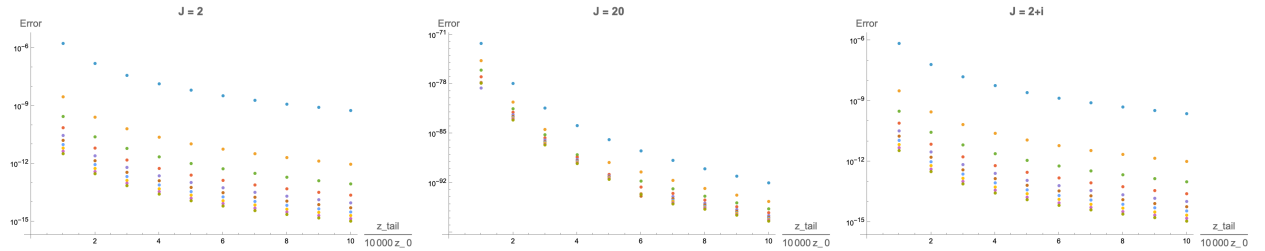


Figure 4.3: The systematic truncation error $\text{Error}_{\text{tail}}$ plotted as a function of the normalized cutoff scale $\frac{z_{\text{tail}}}{10^4 z_1}$ for three distinct values of spin J . Each plot traces ten sampled energy levels distributed between $s = 5$ (blue dots) and $s = 95$ (green dots).

4.2.3 Regularization of Threshold Singularities

Having resolved the numerical challenges associated with the asymptotic tail, we must address a structural divergence arising at the threshold boundary. In $d = 4$, the leading threshold singularity of the amplitude scales at worst as $(z - z_1)^{-1/2}$, which is square-integrable at the lower bound. However, in higher dimensions—as detailed in Section 3.1—the conformal wavelet expansion introduces non-integrable algebraic singularities scaling up to $(z - z_1)^{-\frac{d-3}{2}}$. Consequently, our numerical framework requires a systematic prescription to evaluate master integrals containing boundary divergences of the form $(z - z_1)^{-\frac{\tilde{n}+1}{2}}$, where $\tilde{n} \in \mathbb{N}$.

While these expressions appear pathological if the Froissart-Gribov integral (2.18) is evaluated literally along the real axis, this issue is elegantly bypassed by returning to the underlying complex geometry of the deflation contour. Rather than integrating the isolated real discontinuity, we deform the integration path into the complex z -plane around the branch point at z_1 . If $\tilde{n}$ is odd, the exponent $\frac{\tilde{n}+1}{2} = l \in \mathbb{N}^*$ corresponds to an isolated pole, allowing us to evaluate the localized threshold contribution via a standard contour residue. If $\tilde{n}$ is even, the exponent is a half-integer, defining a branch point that can be evaluated using a keyhole contour deflected into the complex plane, as illustrated in Fig. 4.4.

Since the kinematic kernel $z \mapsto (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z)$ is strictly analytic in the neighborhood of z_1 , executing the residue extraction for the polar configurations is straightforward via Cauchy’s differentiation formula:

$$\oint_{z_1} dz \frac{(z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z)}{(z - z_1)^l} = \frac{2\pi i}{(l-1)!} \frac{d^{l-1}}{dz^{l-1}} \left((z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z) \right) \Big|_{z \rightarrow z_1}, \quad l \in \mathbb{N}^*. \quad (4.14)$$

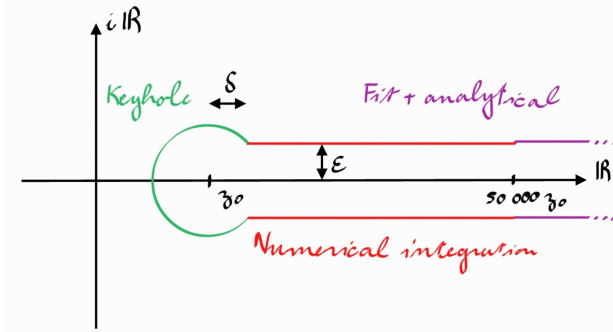


Figure 4.4: Deformed integration contours in the complex z -plane used to regularize threshold singularities during numerical Froissart-Gribov projection.

In the case where $\tilde{n} = 2n$ is even, evaluating the keyhole projection requires careful management of the branch cut. Expanding the analytic part of the integrand to zeroth order around the threshold z_1 , the localized singular contribution near the tip of the contour reduces to the following master integral:

$$\int_{\Gamma_{\text{tip}}} \frac{dz}{(z_1 - z)^{n+\frac{1}{2}}} = \frac{2i(-1)^n}{(\frac{1}{2} - n)(\delta + i\varepsilon)^{n-\frac{1}{2}}}, \quad (4.15)$$

where the geometric regularization parameters ε and δ correspond to the vertical offset and horizontal threshold backstep defined in Fig. 4.4. Scaling this geometric factor by the localized kinematic residue $(z_1^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z_1)$ yields the leading-order threshold contribution.

To quantify the systematic truncation error introduced by this local approximation, we expand the regular kernel $f(z) \equiv (z^2 - 1)^{\frac{d-4}{2}} Q_J^{(d)}(z)$ to first order, $f(z) \approx f(z_1) + f'(z_1)(z - z_1)$. This yields the relative threshold error estimator:

$$\text{Error}_{\text{thresh}} \approx \left| \frac{f'(z_1)}{f(z_1)} (\delta + i\varepsilon) \right| \sim \left| \frac{\partial_z Q_J^{(d)}(z_1)}{Q_J^{(d)}(z_1)} \right| \sqrt{\delta^2 + \varepsilon^2}. \quad (4.16)$$

Consequently, to bound this local truncation error strictly below our global tolerance of 10^{-6} , the horizontal deformation scale δ must be dynamically adapted to track the local regularized scale of the Gegenbauer Q -functions:

$$\delta \approx \left| \frac{Q_J^{(d)}(z_1)}{\partial_z Q_J^{(d)}(z_1)} \right| \times 10^{-6}. \quad (4.17)$$

The remaining smooth contributions over the regularized core domain $[z_1 + \delta, z_{\text{tail}}]$ are evaluated efficiently using standard adaptive numerical quadrature routines. By partitioning the domain and isolating the singular boundaries, the execution time for the full algorithm is optimized to approximately 10 seconds per kinematic point (s, J) . This high-performance numerical pipeline yields the complex angular momentum spectra presented in Section 5.1.

4.3 Tracking Regge Trajectories via Complex Secant Root-Finding

Using the regularized projection framework detailed above, one can systematically evaluate the partial-wave amplitude $f_J(s)$ for any fixed physical energy $s > 4$ across the complex angular momentum plane $J \in \mathbb{C}$ to locate its singularity structure. Due to the multi-sheeted geometry of the underlying S -matrix, the physical sheet evaluations do not expose the unphysical Regge poles directly as divergent peaks; instead, they manifest as localized zeros.¹

While mapping a dense, multi-dimensional numerical grid over $\mathbb{C}$ at sequential values of s can successfully trace these Regge trajectories—mapping the functional evolution $J(s)$ —such brute-force scanning is computationally prohibitive, requiring days of continuous processing. Standard gradient-driven optimization, such as the Newton-Raphson method [12], offers a more targeted alternative but demands the computation of analytical or finite-difference derivatives, which doubles or triples the number of costly numerical integrations required per iteration. To maximize computational efficiency, we implement a complex extension of the secant method [19], structured as follows:

- Initialize the solver with two closely spaced complex configurations J_1 and J_2 to construct a local linear approximation, as sketched in Fig. 4.5.
- Compute the intercept point at which this secant line vanishes, defining the next iterate J_3 .
- Update the search window by shifting the historical data points ($J_2 \rightarrow J_1$ and $J_3 \rightarrow J_2$) and iterate the process.

The iterative solver terminates when the residual amplitude falls below a strict numerical threshold, $|f_{J_3}(s)| < 10^{-10}$, isolating genuine roots from background numerical noise. To safeguard the pipeline against unphysical non-convergent trajectories and localized numerical instabilities, we enforce two strict regularizing constraints: the step must exhibit strict monotonic descent, $|f_{J_3}(s)| < |f_{J_2}(s)|$, and convergence must be achieved within a maximum allocation of 20 iterations.

By implementing an internal cache for the computed basis integrals, the operational cost is reduced to a single forward evaluation per iteration step. Furthermore, to accelerate convergence, we employ a high-order, three-point modified secant method rather than the simple linear variant, following [19]. To trace a complete trajectory continuous in s , we establish a baseline at $s = 5$ and construct a broad scanning profile of the projection (cf. Fig. 5.1). The approximate positions of the zeros are extracted visually and used to initialize the secant pipeline, mapping out the exact complex coordinates of the root. The full trajectory is then swept automatically via a continuation prescription:

¹From elastic unitarity, the S -matrix satisfies $S_{\text{physical}}(J, s)S_{\text{unphysical}}(J, s) = 1$. Consequently, a simple pole residing on the unphysical sheet translates directly into a identifiable zero on the physical sheet projection.

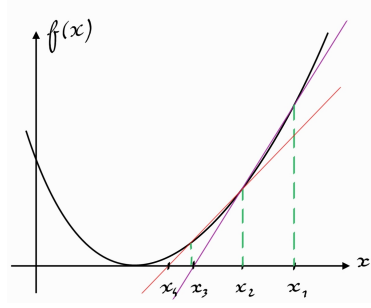


Figure 4.5: Schematic representation of the standard secant root-finding method projected onto a real one-dimensional axis.

- Increment the center-of-mass energy by a small kinematic step ($\Delta s = 0.1$).
- Seed the initialization pair $\{J_1, J_2\}$ for the new energy level using the converged root coordinate discovered at the previous step s .
- Execute the high-order secant algorithm to isolate the updated root position, and repeat the incremental sweep.

This dynamic tracking pipeline maps out the comprehensive trajectories discussed in Detail in [Section 5.2](#).

Chapter 5

Numerical Spectra and Analytical Regge Asymptotics

In this chapter, we present the physical results obtained by applying the numerical projection and tracking framework developed in Chapter 4. These findings map out the complex angular momentum spectra for our structural ansatz, establishing preliminary boundaries on the non-perturbative S-matrix space. Because these calculations represent an ongoing exploration, we contextualize our current numerical datasets alongside ongoing computational sweeps targeting extended amplitude configurations.

Concurrently, Section 5.3 provides a self-contained analysis of a parallel line of inquiry originating from our prior investigation into the exact analytic properties of the Veneziano amplitude. This study complements the exploration of higher-dimensional behavior in UV-complete theories by providing explicit analytical results.

5.1 Exploring the complex J plane

Using the methods described in section 4.2, we can create plots representing $f_J(s)$ for different amplitudes. The computation takes about 4 hours to get plots like the one showed in figure 5.1. These are created with a coarse grid (spacing 0.2) refined on 10% of cells where the variation is most important.

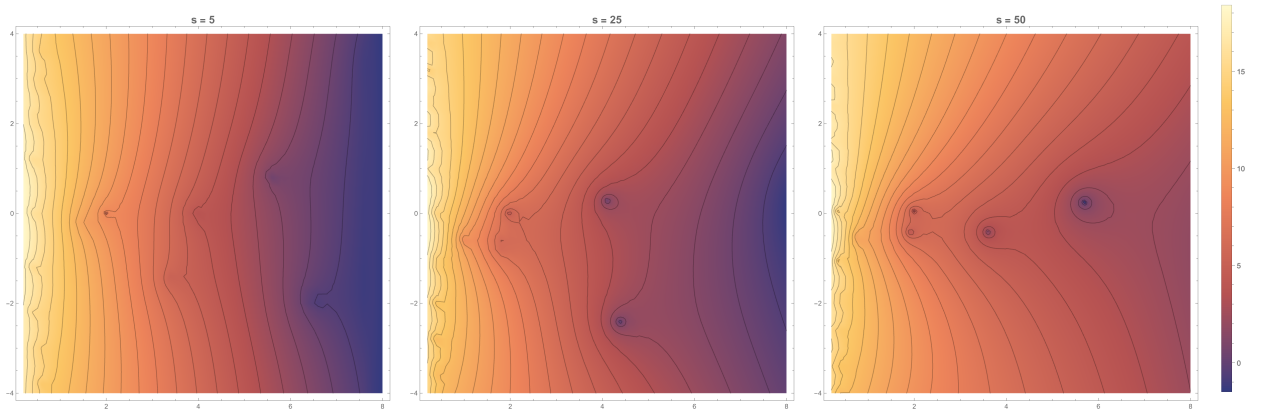


Figure 5.1: Image of $\log |f_J(s)|$ for $d = 6$, $c_{0,\max}$, $N_{\max} = 20$ (see [Simon's website](#)).

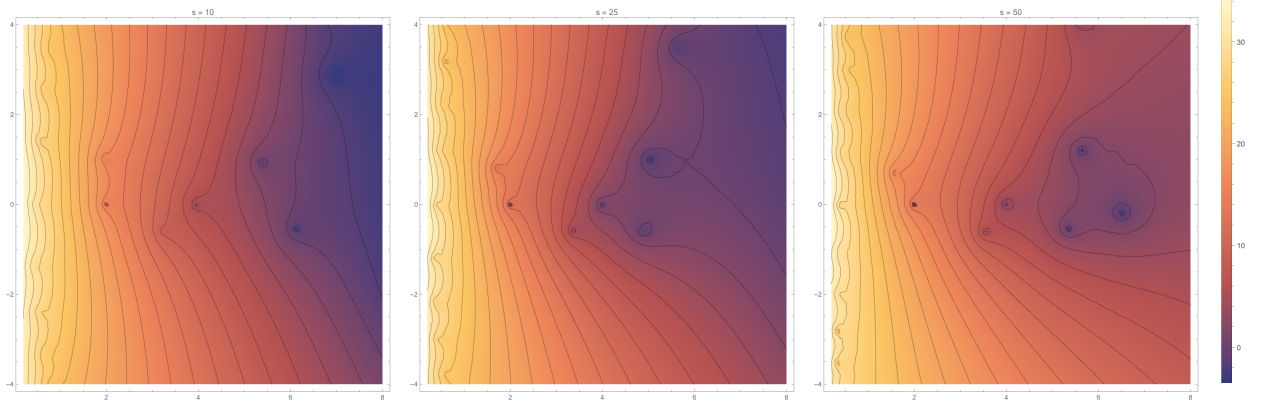


Figure 5.2: Image of $\log |f_J(s)|$ for $d = 4$, $c_{0,\min}$, $N_{\max} = 27$ (see [Simon's website](#)).

In both figure 5.1 and 5.2 we can observe zeroes which will be monitored more closely in the next section. We can also observe what may be a branch cut around $\Re(J) = 0$. This structure might be nonphysical and further computation, especially with different size of ansatz, are planned for the last two week of the internship in order to explore the origin of these structures.

5.2 Evolution of the Regge Trajectories

Using the local minima isolated from the initial complex landscape at $s = 5$ (cf. Fig. 5.1) to seed our continuation algorithm, we track the dynamic evolution of the roots across the complex J -plane to map the Regge trajectories $J(s)$ for $d = 6$, $c_{0,\max}$, and $N_{\max} = 20$. Based on the non-perturbative bootstrap bounds derived in [8] and visualizable on the [SDPB Wrap Display portal](#), the standard spectral framework predicts a single isolated physical Regge trajectory. However, our complex numerical tracking uncovers a more intricate structure, yielding three distinct trajectories as illustrated in Fig. 5.3.

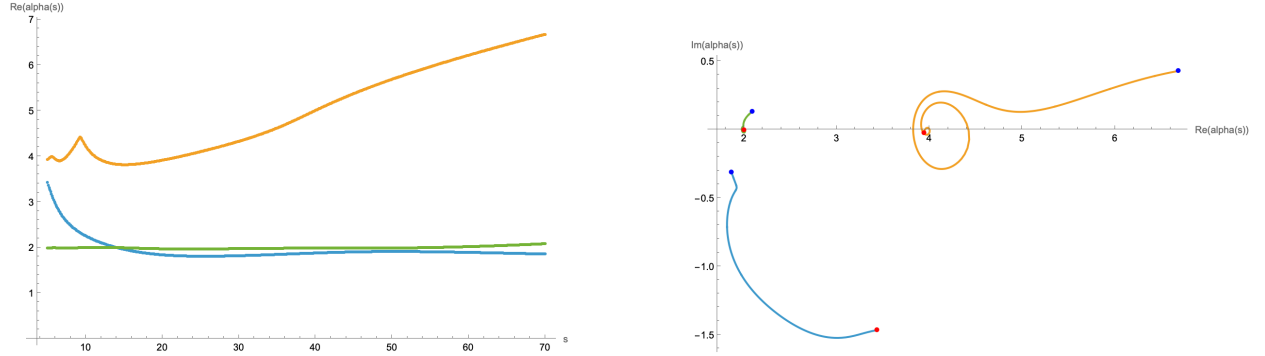


Figure 5.3: Dynamic Regge trajectories in the complex J -plane for $d = 6$, $c_{0,\max}$, and $N_{\max} = 20$. On the left panel, the red marker denotes the baseline configuration at $s = 5$, while the blue marker tracks the high-energy limit at $s = 70$.

The trajectory highlighted in orange exhibits some structural agreement with the analytical predictions of Gumus et al. for the high-energy regime ($s > 30$). Conversely, the low-energy onset of this curve, alongside the two remaining trajectories, displays highly anomalous, migratory paths that appear unphysical.

A closely related phenomenon involving trajectory discontinuities has been reported in alternative kinematic setups; specifically, Correia et al. [4] observed sharp tracking discontinuities when mapping roots in the complex s -plane at fixed values of J . In their framework, the physical trajectory appears to terminate or jump because the underlying pole migrates across the physical s -channel branch cut, escaping onto an unphysical Riemann sheet. Crucially, this masking effect was shown to systematically recede as the truncation cutoff $N_{\max}$ was increased.

In our current fixed- s tracking architecture, the complex variable is $J \in \mathbb{C}$. Because the Froissart-Gribov kernel does not possess an analogous dynamic branch cut in the angular momentum plane, the tracking root cannot smoothly escape onto an adjacent sheet; instead, it is forced to remain on the evaluation plane, where it manifests as an unphysical, highly erratic trajectory. To rigorously decouple these truncation artifacts from genuine non-perturbative features, systematic finite-size scaling analyses across varying ansatz depths are required. In particular, a targeted investigation of the $d = 4$, $c_{0,\min}$ configuration—which provides a direct point of comparison to the structures identified in [4]—constitutes a primary objective for the conclusion of this work.

5.3 Analytical Regge Structure and Residues of the Veneziano Amplitude

To complete the study of higher dimension theories, we take a look at the Veneziano amplitude across dimension.

5.3.1 The Regge Residues of the Veneziano Amplitude

The formulation of the Veneziano amplitude represents a milestone in modern mathematical physics, serving as the historical catalyst for the development of string theory and offering a concrete realization of UV-complete, infinitely rising Regge trajectories. In Mandelstam variables, the dual-resonance scattering amplitude for tachyonic scalar states is defined via the Euler Beta function:

$$A(s, t) = B(-s - 1, -t - 1) = \int_0^1 du u^{-s-2} (1-u)^{-t-2}. \quad (5.1)$$

By construction, this representation manifestly satisfies $s \leftrightarrow t$ crossing symmetry and possesses a meromorphic singularity structure poles restricted entirely to the real axis. Furthermore, the modern deployment of the no-ghost theorem guarantees that the spectrum is free of unphysical negative-norm states, preserving strict tree-level unitarity provided the spacetime dimension satisfies $d \leq 10$.

A defining hallmark of the Veneziano amplitude is its clean representation in terms of linear Regge trajectories. Upon performing a partial-wave projection, the amplitude $f_J(s)$ can be decomposed into an infinite sequence of poles in the complex angular momentum plane, tracking both the leading trajectory and its corresponding daughter trajectories:

$$f_J(s) = \sum_{k=0}^{+\infty} \frac{\tilde{\beta}_k(J, s)}{J - (s - k)}, \quad (5.2)$$

where $\tilde{\beta}_k(J, s)$ denotes the generalized Regge residue associated with the k -th daughter trajectory.

From this expression, we can isolate the on-shell partial-wave Regge residues by evaluating the off-shell amplitude directly on the pole trajectory, defining $\beta_k(s) \equiv \tilde{\beta}_k(s - k, s)$. Internal consistency and tree-level unitarity dictate that these residues remain strictly positive across the physical kinematic domain $s > 0$ [21]. Utilizing the algebraic framework outlined in Appendix A.1, the closed-form expression for these residues reads:

$$\beta_k(s) = \beta_0(s) \frac{(s - k + \frac{d-1}{2} + \lfloor \frac{k}{2} \rfloor)_{\lfloor \frac{k+1}{2} \rfloor} p_k(s)}{s^k}, \quad (5.3)$$

where the overall scale factor is governed by the ground-state residue,

$$\beta_0(s) = \frac{\pi^{\frac{3-d}{2}} s^s}{2^{2s+2d-4} \Gamma\left(s + \frac{d-1}{2}\right)}, \quad (5.4)$$

and the structural polynomials $p_k(s)$ are defined via a weighted sum over higher-order Nörlund generalized Bernoulli polynomials $B_n^{(\alpha)}(x)$ [17]:

$$p_k(s) = (-1)^k 2^{2k} \sum_{n=0}^{\lfloor \frac{k}{2} \rfloor} \frac{(s - k + \frac{d-1}{2} + n)_{\lfloor \frac{k}{2} \rfloor - n}}{2^{4n} n! (k - 2n)!} (s)^{2n} B_{k-2n}^{(s+1)}\left(\frac{s}{2}\right). \quad (5.5)$$

To analyze the positivity constraints more transparently, $p_k(s+k)$ can be mapped onto an multi-dimensional integral representation. For the forward kinematic region $s > 0$, unitarity is preserved if and only if $p_k(s+k) > 0$. In our subsequent asymptotic analysis, we employ the following compact contour-integral representation:

$$p_k(s+k) = \frac{(-1)^k 2^{2k}}{\sqrt{\pi} k!} \frac{\Gamma\left(s + \frac{d-1}{2}\right)}{\Gamma\left(s + \frac{d-2}{2}\right)} \left(s + \frac{d-1}{2}\right) \oint_{C_0} \frac{dt}{t^{k+1}} \int_{-1}^1 du (1-u^2)^{s+\frac{d-4}{2}} e^{\frac{s+k}{2}(1+u)t} \left(\frac{t}{e^t - 1}\right)^{s+k+1}, \quad (5.6)$$

where C_0 denotes a closed contour in the complex t -plane encircling the origin.

5.3.2 Asymptotic Distribution of Zero Attractors

In preparing these algebraic expressions for publication, a systematic numerical exploration of the roots of $p_k(k(z+1))$ in the complex z -plane revealed that the zeros do not distribute randomly; rather, they accumulate cleanly along a sharp, well-defined limit curve in the large- k limit, as illustrated in Fig. 5.4.

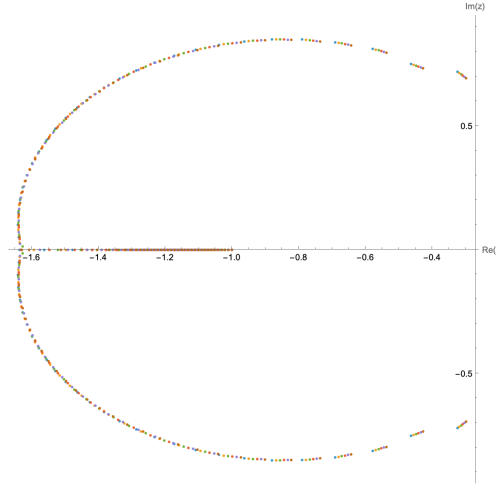


Figure 5.4: Numerical root distribution of the structural polynomials $p_k(k(z+1))$ for $d = 0$ across the overlapping degree interval $90 \leq k \leq 100$.

The emergence of such geometric loci is a classic hallmark of the Stokes phenomenon, where the zero-attractor curves correspond to the curves of equal real action (anti-Stokes lines) connecting competing saddle points of the integral representation (see Appendix A.2 for a foundational toy model, or [13] for an explicit Riemann-Hilbert analysis of related orthogonal systems). To analyze this behavior via a semiclassical

expansion, we isolate the large- k scaling parameter by recasting the multi-dimensional integral into an exponential form:

$$p_k(k(z+1)) \propto \frac{1}{2\pi i} \oint_{C_0} dt \int_{-1}^1 du \exp[k\Phi(t, u; z, k)] , \quad (5.7)$$

where the multi-variable effective action $\Phi(t, u; z, k)$ is given explicitly by:

$$\Phi(t, u; z, k) = -\left(\frac{1}{k} + z + 1\right) \log(e^t - 1) + \frac{1}{2}t(u+1)(z+1) + z \log(t) + z \log(1-u^2) . \quad (5.8)$$

5.3.3 Numerical Saddle-Point Analysis and the Stokes Landscape

By solving the algebraic saddle-point condition for the internal coordinate, $\partial_u \Phi = 0$, we map the system onto two distinct topological branches in the u -plane. Injecting these configurations back into the master effective action yields two corresponding semi-integrated actions:

$$\tilde{\Phi}_-(t; z, k) = \frac{1}{2} \left[-\frac{2(kz + k + 1) \log(e^t - 1)}{k} - \sqrt{t^2(z+1)^2 + 4z^2} \right. \quad (5.9)$$

$$\left. + 2z \left(\log \left(-\frac{4z}{\sqrt{t^2(z+1)^2 + 4z^2} - 2z} \right) + \log(t) - 1 \right) + t(z+1) \right] , \quad (5.10)$$

$$\tilde{\Phi}_+(t; z, k) = \frac{1}{2} \left[-\frac{2(kz + k + 1) \log(e^t - 1)}{k} + \sqrt{t^2(z+1)^2 + 4z^2} \right. \quad (5.11)$$

$$\left. + 2z \left(\log \left(\frac{4z}{\sqrt{t^2(z+1)^2 + 4z^2} + 2z} \right) + \log(t) - 1 \right) + t(z+1) \right] . \quad (5.12)$$

Differentiating these localized actions with respect to the remaining complex coordinates leads to a dual system of transcendental saddle-point equations for the parameter $y \equiv \frac{t}{2}$:

$$-\sqrt{y^2(z+1)^2 + z^2} - \frac{y[(kz + k + 1) \coth(y) + 1]}{k} = 0 , \quad (5.13)$$

$$\sqrt{y^2(z+1)^2 + z^2} - \frac{y[(kz + k + 1) \coth(y) + 1]}{k} = 0 . \quad (5.14)$$

Unlike the simplified framework evaluated in Appendix A.2, this system resists closed-form inversion. To visualize the global configuration of all critical points simultaneously for a fixed choice of z without encountering Riemann sheet ambiguities, we construct a consolidated master function $\hat{\Phi}(y; z, k)$ by taking the product of Eq. (5.13) and Eq. (5.14):

$$\hat{\Phi}(y; z, k) = \frac{-k^2 [y^2(z+1)^2 + z^2] + y^2(kz + k + 1)^2 \coth^2(y) + 2y^2(kz + k + 1) \coth(y) + y^2}{k^2} . \quad (5.15)$$

The roots of this function, mapping the exact positions of the complex saddles, are plotted in Fig. 5.5.

The numerical scan reveals an infinite, discretely spacing tower of saddle points. In the asymptotic limit, the dominant contributions are governed by the primary low-action roots. Restricting the asymptotic expansion to the first two leading saddles yields the dominant phase boundaries illustrated in Fig. 5.6. While this two-saddle truncation successfully reproduces the zero-attractor trajectories observed along the real axis, it fails to capture the auxiliary geometric branches. Preliminary multi-saddle tests indicate that a six-saddle configuration is required to fully resolve the non-trivial topology of the limit curves—a derivation that constitutes a primary focus for the final phase of this investigation.

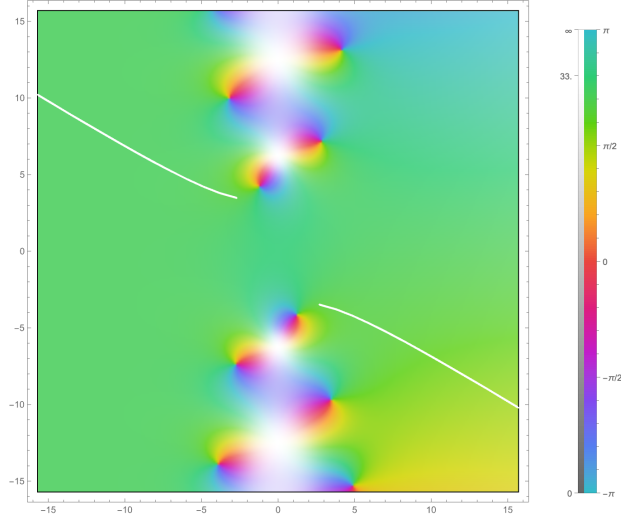


Figure 5.5: Global landscape of the master saddle-point function $\hat{\Phi}(y; z, k)$ evaluated at $k = 100$ and $z = -1.4 + 0.64i$ within the complex y -plane.

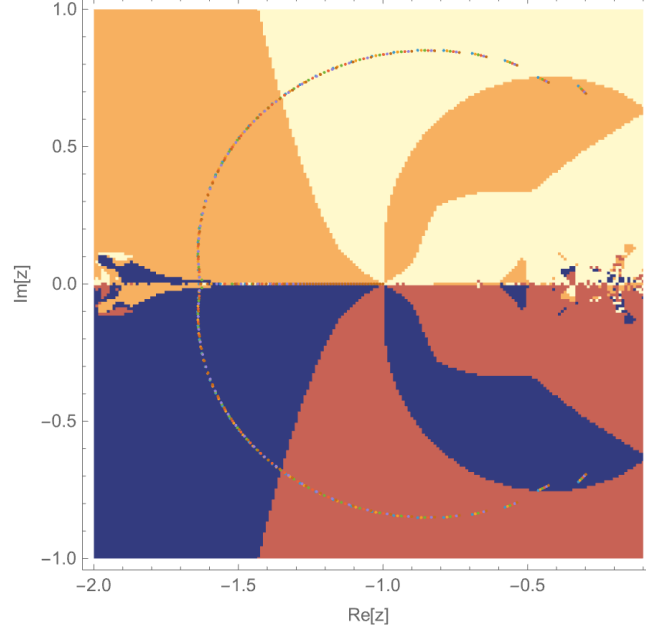


Figure 5.6: Dominant exponential phase domains (Stokes/anti-Stokes regions) computed by truncating the system to the first two primary saddles. The four distinct color bands map the competitive dominance boundaries between the two distinct saddles across both structural action branches $\tilde{\Phi}_{\pm}$.

Chapter 6

Conclusion and Outlook

6.1 Academic and Methodological Developments

Following a comprehensive literature review of the primal S-matrix bootstrap program—with a particular focus on the non-perturbative bounds established in [8]—the primary academic focus of this internship centered on the mathematical structures underlying the Froissart-Gribov projection formula. Building upon our prior analytical explorations of the Veneziano amplitude, the simultaneous deployment of analytical expansions and numerical tracking schemes within the ansatz framework of Section 3.1 provided a significantly more rigorous understanding of the complex angular momentum landscape. In particular, mapping the non-trivial analytic structure of the kinematic kernel required navigating the geometry of Riemann surfaces and sheet-switching mechanisms; while these features ultimately acted as boundary constraints rather than dynamical conduits for our specific numerical runs, mastering their global topology proved essential for interpreting our tracking data.

Although the derivation of a general closed-form analytical solution for the projection of our finite-size wavelets encountered fundamental algebraic limits, the systematic exploration of the complex landscape catalyzed the development of a highly efficient numerical architecture for evaluating the Froissart-Gribov integral, as detailed in Section 4.2. By coupling this optimized integration routine with an adaptive secant-method continuation algorithm, we formulated a novel pipeline to dynamically map complex Regge trajectories $J(s)$ across continuous energy intervals (cf. Section 4.3). Utilizing these original computational tools, we initiated a systematic study of the optimal amplitudes isolated by the SDPB solver in [8]. The resulting phenomena, documented in Sections 5.1 and 5.2, uncovered intricate structural variations and multi-branched tracking profiles. To firmly decouple genuine non-perturbative physics from truncation-induced artifacts, expanding our dataset via alternative amplitude configurations and deeper ansatz spaces remains an active, ongoing computational objective.

In parallel with our bootstrap investigations, we extended our previous analytical derivations regarding the on-shell Regge residues $\beta_k(s)$ of the open bosonic string tree amplitude. This line of research required a deep dive into the semiclassical mechanics governing the asymptotic accumulation of polynomial zeros along sharp geometric limit curves, an evaluation deeply intertwined with multi-dimensional Picard-Lefschetz theory and the Stokes phenomenon. While these zero-attractor trajectories were successfully mapped and understood analytically for the single-variable confluent hypergeometrics evaluated in our Appendix A.2 toy model, resolving the complex phase boundaries of the highly transcendental Nörlund-type polynomials of interest remains an active area of research. The complete structural classification of these multi-saddle limit curves is currently being finalized and is slated to be part of a forthcoming peer-reviewed publication.

6.2 Personal and Professional Development

On a personal and professional level, this research internship has been an immensely enriching experience, marking a significant evolution in my methodological perspective. From a technical standpoint, the implementation of advanced numerical integration schemes and automated continuation algorithms represented a profound shift in my approach to mathematical physics. Prior to this work, I viewed numerical simulation primarily as a heuristic precursor—a tool restricted to building initial intuition for complex landscapes with the ultimate goal of translating those observations into exact analytical proofs. However, directly grappling with the constraints of the S-matrix bootstrap has broadened this philosophy. I now more fully appreciate that high-performance numerical architectures are not merely secondary approximations, but foundational engines capable of driving the bootstrap program forward and uncovering non-perturbative features that may entirely elude closed-form representations.

In addition to the core research, active participation in the diverse seminars, group meetings, and academic presentations hosted at LAPTh provided a vital window into the broader theoretical physics landscape. Exposure to these varied research vectors proved exceptionally timely as I prepare to select my advanced coursework for the Master of Advanced Study (MASt) Part III program at the University of Cambridge. Engaging with this vibrant intellectual community also underscored the deeply collaborative nature of modern research, highlighting the human exchanges and collective problem-solving dynamics that sustain and enrich the academic ecosystem.

Appendix A

On the Veneziano amplitude

A.1 Deriving the Regge residue of the Veneziano amplitude

In order to study those poles, I will follow the steps of article [6]. The first thing to do is to approximate the integral in 2.18. In order to do so, we need to simplify it by removing the kernel. This can be done using this relation (see DLMF 15.8.12):

$${}_2F_1\left(\frac{J+d-3}{2}, \frac{J+d-2}{2}, J + \frac{d-1}{2}, \frac{1}{z^2}\right) = \left(1 - \frac{1}{z^2}\right)^{-\frac{d-4}{2}} {}_2F_1\left(\frac{J+1}{2}, \frac{J+2}{2}, J + \frac{d-1}{2}, \frac{1}{z^2}\right) \quad (\text{A.1})$$

The equation 2.18 now has the form:

$$f_J(s) = \frac{2\mathcal{N}_d c_J^{(d)}}{\pi} \int_{z_1}^{+\infty} \frac{dz}{z^{J+1}} {}_2F_1\left(\frac{J+1}{2}, \frac{J+2}{2}, J + \frac{d-1}{2}, \frac{1}{z^2}\right) \text{Disc}_t[A(s, z)] \quad (\text{A.2})$$

We are now going to get closer to the Veneziano amplitude by putting $z_1 = 1 + \frac{2}{s}$ and supposing a development of the form $\text{Disc}_t[A(s, z)] = z^s \sum_n a_n(s) z^{-n}$. Using the known expression of ${}_2F_1$, we arrive at the following form:

$$f_J(s) \approx \sum_{k=0}^{+\infty} \int_{1+\frac{2}{s}}^{+\infty} dz z^{s-J-k-1} \tilde{\beta}_k(J, s) \quad (\text{A.3})$$

$$= \sum_{k=0}^{+\infty} \frac{\tilde{\beta}_k(J, s)}{J - (s - k)} \quad (\text{A.4})$$

We can identify $\alpha_k(s) = s - k$ in this case and $\beta(s) = \tilde{\beta}(s - k, s)$. We want to compute the latter, this means that we are going to need an expression for the asymptotic development of $\text{Disc}_t[A(s, z)]$.

The expression of the Veneziano amplitude is given in equation 5.1. It is possible to make the discontinuity apparent using the inversion formula of the Euler gamma function. This gives us:

$$\text{Disc}_t[B(-s, -t(s, z))] = \frac{\pi \Gamma(s + t + 1)}{\Gamma(s + 1) \Gamma(t + 1)} \quad (\text{A.5})$$

It is possible to reorganize this to get back a Euler beta function which is defined by an integral. We use the Laplace method to compute the asymptotical development. This method is based on the following form of integrals:

$$I(z) = \int_0^{+\infty} du e^{-zu} f(u) \quad (\text{A.6})$$

Then, if $f(u)$ is analytic and with the following development : $f(u) = \sum_{n=0}^{+\infty} \frac{a_n}{n!} u^n$. Then the integral has this asymptotic development (see [18] for more details):

$$I(z) \approx \sum_{n=0}^{+\infty} \frac{a_n}{z^{n+1}} \quad (\text{A.7})$$

Using the integral representation of the Beta function I managed to compute the following development:

$$\text{Disc}_t[B(-s, -t(s, z))] = \frac{\pi \Gamma(s \frac{z+1}{2})}{\Gamma(s+1) \Gamma(s \frac{z-1}{2})} \quad (\text{A.8})$$

$$\approx \frac{\pi \left(\frac{zs}{2}\right)^s}{\Gamma(s+1)} \sum_{n=0}^{+\infty} \frac{B_n^{(s+1)} \left(\frac{s}{2}\right)}{n!} (-s)_n \left(\frac{zs}{2}\right)^{-n} \quad (\text{A.9})$$

With this development, we can start to compute the Regge poles. First, we need the development of ${}_2F_1$:

$${}_2F_1\left(\frac{J+1}{2}, \frac{J+2}{2}, J + \frac{d-1}{2}, \frac{1}{z^2}\right) = \sum_{n=0}^{+\infty} \frac{(J+1)_{2n}}{\left(J + \frac{d-1}{2}\right)_n 2^{2n} n!} z^{-2n} \quad (\text{A.10})$$

Where $(a)_n = \prod_{l=0}^{n-1} (a+l) = \frac{\Gamma(a+n)}{\Gamma(a)}$ is the Pochhammer symbol. From these two developments, it is obvious that:

$$\tilde{\beta}_k(J, s) = \frac{2\mathcal{N}_d c_J^{(d)}}{\pi} \frac{\pi \left(\frac{s}{2}\right)^s}{\Gamma(s+1)} \sum_{n=0}^{\lfloor \frac{k}{2} \rfloor} \frac{(J+1)_{2n}}{\left(J + \frac{d-1}{2}\right)_n 2^{2n} n!} \frac{B_{k-2n}^{(s+1)} \left(\frac{s}{2}\right)}{(k-2n)!} (-s)_{k-2n} \left(\frac{s}{2}\right)^{-(k-2n)} \quad (\text{A.11})$$

It is now time to use $\beta_k(s) = \tilde{\beta}_k(s-k, s)$:

$$\beta_k(s) = \beta_0(s) \frac{\left(s-k + \frac{d-1}{2} + \lfloor \frac{k}{2} \rfloor\right)_{\lfloor \frac{k+1}{2} \rfloor} p_k(s)}{s^k} \quad (\text{A.12})$$

Where $\beta_0(s) = \frac{\pi^{\frac{3-d}{2}} s^s}{2^{2s+2d-4} \Gamma(s + \frac{d-1}{2})}$ which is clearly positive for $s \geq 0$ and:

$$p_k(s) = (-1)^k 2^{2k} \sum_{n=0}^{\lfloor \frac{k}{2} \rfloor} \frac{\left(s-k + \frac{d-1}{2} + n\right)_{\lfloor \frac{k}{2} \rfloor - n}}{2^{4n} n! (k-2n)!} (s)^{2n} B_{k-2n}^{(s+1)} \left(\frac{s}{2}\right) \quad (\text{A.13})$$

A.2 A Toy Model for Root Accumulation: The Generalized Bessel Polynomials

To illustrate the geometric stabilization of zero-attractor curves through the prism of semiclassical analysis, we examine the generalized Bessel polynomials. These objects admit a hypergeometric definition that can

be mapped directly onto an integral representation:

$$y_n\left(\frac{z}{n}\right) = {}_2F_0\left(-n, n; -\frac{z}{2n}\right) \quad (\text{A.14})$$

$$= \frac{1}{\Gamma(n)} \int_0^{+\infty} du \exp[n\phi(u; z, n)] , \quad (\text{A.15})$$

where the variable-dependent effective action $\phi(u; z, n)$ isolates the large- n scaling behaviors:

$$\phi(u; z, n) = \log\left(1 + \frac{uz}{2n}\right) - \frac{u}{n} + \left(1 - \frac{2}{n}\right) \log(u) . \quad (\text{A.16})$$

In the asymptotic limit $n \rightarrow +\infty$, the integral is dominated by the critical points satisfying the variational condition $\partial_u \phi = 0$. This yields a quadratic constraint in the integration variable, producing two distinct localized saddle solutions, $u_+(z)$ and $u_-(z)$. By deforming the integration path onto the saddles, the global polynomial can be asymptotically approximated as:

$$y_n\left(\frac{z}{n}\right) \approx \frac{1}{\Gamma(n)} \left[a_+(z) e^{n\phi(u_+(z); z)} + a_-(z) e^{n\phi(u_-(z); z)} \right] , \quad (\text{A.17})$$

where $a_{\pm}(z) \in \mathbb{C}$.

The global landscape of the complex z -plane is consequently partitioned by the relative magnitudes of these exponential profiles. Within domains where $\Re[\phi(u_+(z); z)] > \Re[\phi(u_-(z); z)]$, the first saddle-point contribution is exponentially dominant, completely masking the secondary branch; dually, the reverse hierarchy holds when the inequality flips. Because a single isolated exponential phase cannot vanish, macroscopic zeros can only accumulate where these competing configurations achieve exact asymptotic equilibrium:

$$\Re[\phi(u_+(z); z)] = \Re[\phi(u_-(z); z)] . \quad (\text{A.18})$$

Equation (A.18) defines the anti-Stokes lines of the system, which act as strict geometric attractors for the polynomial roots in the continuum limit [1]. This exact topological mapping is verified numerically in Fig. A.1.

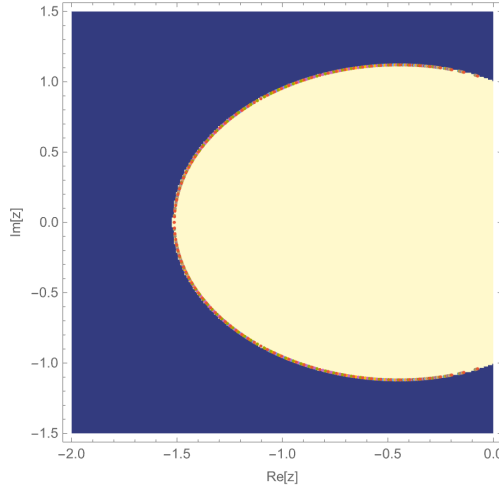


Figure A.1: Phase dominance domains in the complex z -plane for $n = 100$, overlaid with the explicit numerical root distribution of the generalized Bessel polynomials $y_n(z/n)$ across the tracking index interval $90 \leq n \leq 100$.

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